

Networking equipment

Almost all networking equipment is made up of embedded systems, and the primary reason for this is the need to be as fast and reliable as possible. One of the biggest challenges here is the ability to compute the path (for routing). Most of the data-link layer responsibilities are taken care of by the individual port adapters and they don't put any significant load on the system. However, the route calculation can be done on dedicated hardware which speeds up the packet delivery by orders of magnitude. Small Wi-Fi routers in home set-ups use embedded chips for the work. The challenge with dedicated hardware is upgradability and creation of routing logic on hardware. This is more so because there are multiple routing protocols involved and each have their own way of getting the job done.

When complex topologies are to be handled (on higher-end routers), it becomes a challenge to design a chip which can get the job done. Also, the router has to be programmable and the system as a whole needs to run an OS which is familiar to the network administrator (in most cases, this ends up being Linux). This complicates the design because the system would then have to incorporate a more generic processor which can run the OS. The closer the routing logic to hardware, the better and faster the router works.

Inter-planetary systems

Voyager 1 is the first man-made object to leave the solar system and enter inter-stellar space. It has been sending information to NASA scientists about the solar system for the past 35 years! Mars rovers are more examples of ultra-rare systems that are working in an entirely different planet, but controlled from Earth. Satellites of all types are filling our lives with more information and ease than ever before. These systems are short on power. While you might get sunburnt on Earth, you would freeze to death in the low temperatures beyond the asteroid belt. The fact is that the average maximum temperature of Mars is about 27 degrees Celsius, and minimum temperatures go down to less than negative 100 degrees Celsius! Satellites have to deal with a much higher variation in temperatures, since they face much higher maximum temperatures when the sun shines on them.

Most day-to-day communication devices have operational temperature limits because extreme temperatures can damage the equipment. At the same time, power requirements of interplanetary systems are higher than any other wireless device due to sheer distance they have to communicate across (measured in thousands of miles). There are basically two requirements for an interplanetary or extra-planetary system: